Practice workbook · Read two values from a graph, chart or table and find the difference

Two points, bars, peaks and troughs, or ends of a range are located and the gap between them is the answer.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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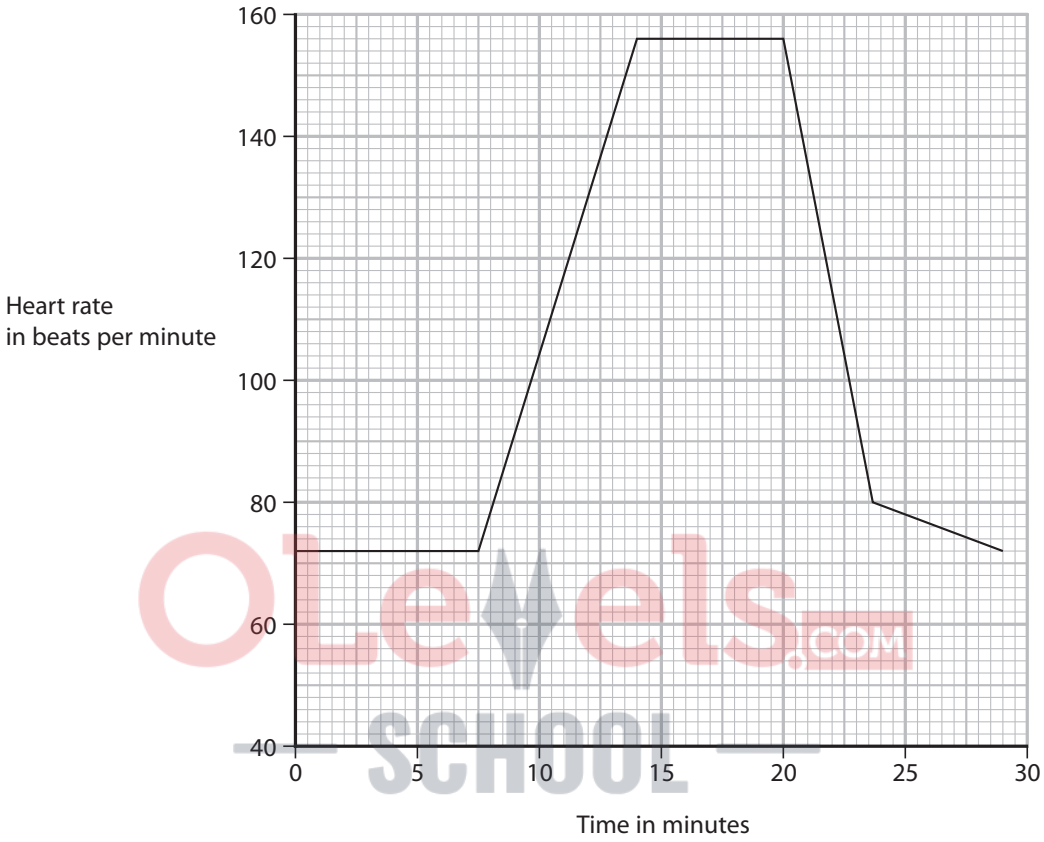
Type ID: HBI-MATH-READ-AND-SUBTRACT

# 1. Question 2 · part a.i · 2 marks

June 2022 Paper 1

Worked-example retry · Attempt part a.i only.

**2** (a) The graph shows how a person's heart rate changes with exercise.




(i) Determine the difference between the maximum heart rate and the resting heart rate.

**(2)**

difference = ..... beats per minute

(ii) Which of these is the time taken for this person to recover from the exercise?

**(1)**

- ☒ **A** 4 minutes
- ☒ **B** 6 minutes
- ☒ **C** 9 minutes
- ☒ **D** 15 minutes

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## 1. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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## 2. Question 5 · part b · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

**5** A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

| Temperature of room in °C | Volume of sweat produced in arbitrary units | |
|---|---|---|
| | Sitting | Active |
| 20 | 1 | 2 |
| 22 | 2 | 6 |
| 24 | 2 | 8 |
| 26 | 2 | 11 |
| 28 | 3 | 14 |
| 30 | 4 | 22 |

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## 2. Question 5 · part b · 2 marks

June 2019 Paper 1R

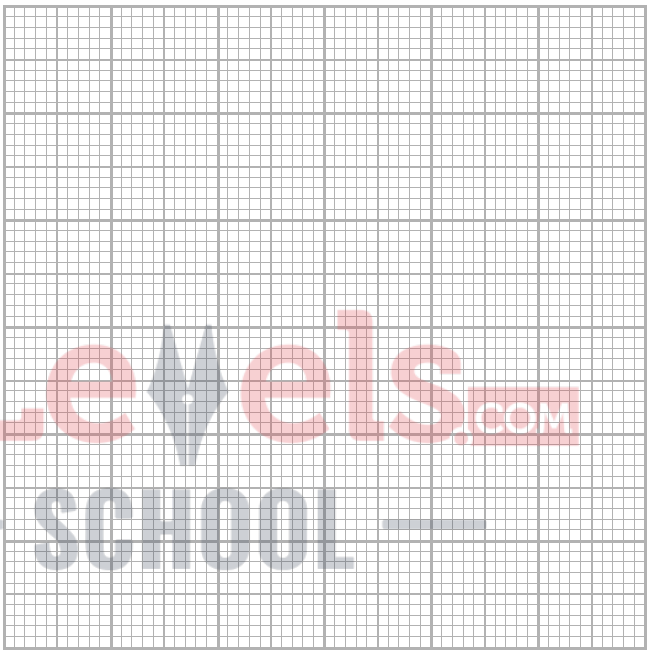
Further practice · Attempt part b only.

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(a) Plot the student's results on the grid.

Join the points with straight lines.

**(5)**



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(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

**(2)**

difference = ..... arbitrary units

## 2. Question 5 · part b · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

(c) Explain the difference between the volume of sweat produced when sitting and when active.

**(3)**

**(Total for Question 5 = 10 marks)**

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Working and answer

## 3. Question 4 · part b.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

**4** (a) State three methods of birth control.

**(3)**

1 .....

.....

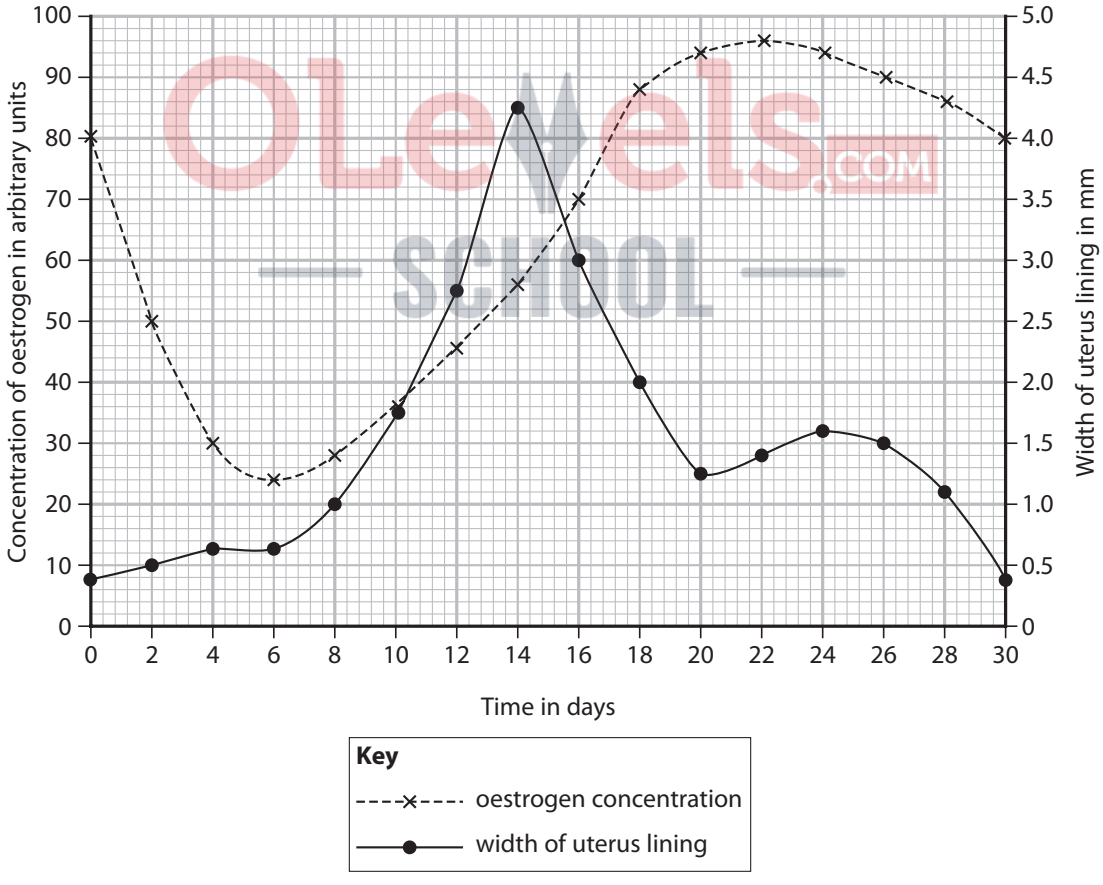
2 .....

.....

3 .....

.....

(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.




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## 3. Question 4 · part b.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part b.i only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

**(2)**

change in width = ..... mm

(ii) Explain the importance of this change in width.

**(3)**

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(iii) Determine during which days menstruation occurred.

**(1)**

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

**(3)**

**(Total for Question 4 = 12 marks)**

## 3. Question 4 · part b.i · 2 marks · Working and answer

Working and answer

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# 4. Question 6 · part a.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

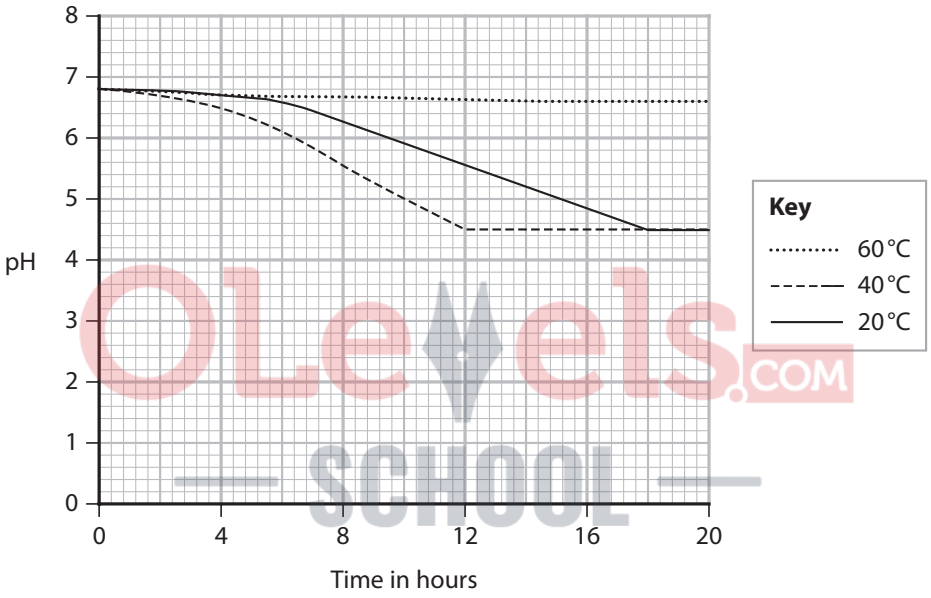
**6** Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.




(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

**(2)**

time = ..... hours

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# 4. Question 6 · part a.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.i only.

(ii) Give two variables that need to be controlled in this investigation.

**(2)**

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

**(2)**

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(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

**(2)**

boiling

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cooling

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## 4. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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# 5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

**6** (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

In 2012, Shinya Yamanaka, a Japanese research scientist, won a Nobel Prize for his work on stem cells. He demonstrated that it was possible to take a sample of blood or skin cells from a person and induce these cells to have the properties of stem cells.

In the past, stem cell research scientists needed to obtain human embryonic stem cells from embryos produced by IVF. This required the agreement of those people involved. Embryonic cells can continually divide and can become any human body cell. Scientists have been exploring the use of embryonic stem cells as medical treatments for many different diseases and conditions.

Using the work of Yamanaka and his team, scientists can now produce cells called induced pluripotent stem cells (iPS cells). Cells from the patient are treated with four factors which control the expression of genes. This produces iPS cells from the patient's own cells, which can then be used in stem cell therapies.

This technique is easy to replicate but there are concerns that unlicensed clinics are offering stem cell treatments. Many of these treatments are expensive and success rates are not proven. In addition, stem cells have the potential to continue to divide in the body. Many of the diseases and conditions that stem cell therapies could be used to treat have no other effective treatments. This makes stem cell therapy a desirable option.

(i) Describe the benefits of producing iPS cells from a patient's own cells.

**(3)**

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## 5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) Give two possible risks of using stem cells to treat diseases and medical conditions.

**(2)**

1 .....

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2 .....

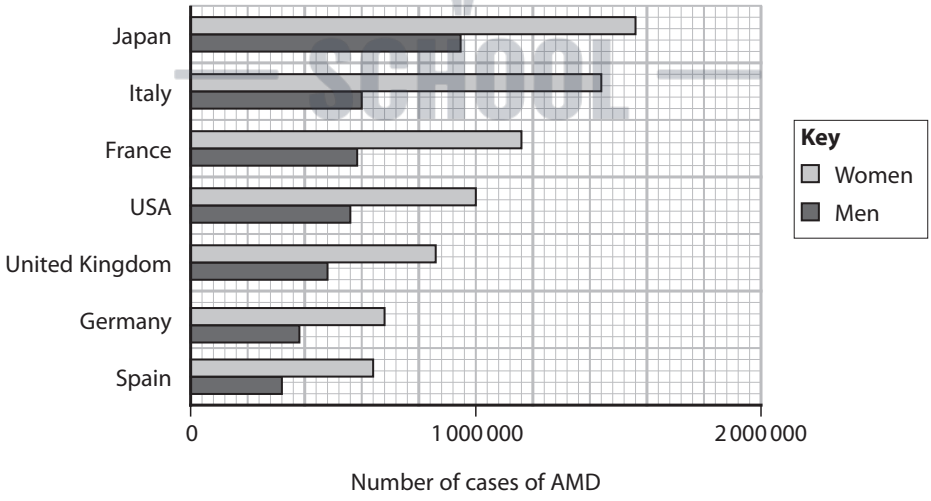
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(b) One condition that stem cells could be used to treat is age-related macular degeneration (AMD).

This is the main cause of vision loss in people over the age of 50.

The graph shows the number of people affected by AMD in some countries.




(i) Calculate the difference between the number of women and the number of men affected by AMD in the United Kingdom.

**(2)**

difference = .....

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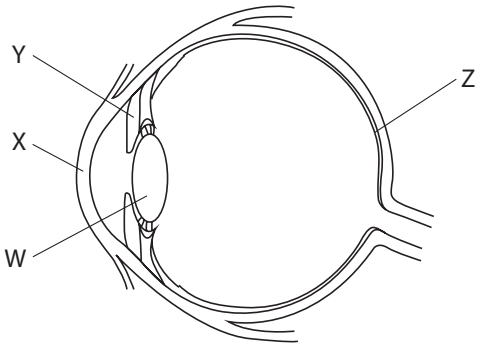
# 5. Question 6 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) The diagram shows a human eye.

AMD affects the retina of the eye.




Which of these structures is affected by AMD?

(1)

- ☒ **A** structure W
- ☒ **B** structure X
- ☒ **C** structure Y
- ☒ **D** structure Z

(c) A deficiency of vitamin A can also cause blindness.

A genetically engineered rice product can be used to prevent deficiency of vitamin A.

Describe how genetic engineering is used to produce this rice product.

(3)

**(Total for Question 6 = 11 marks)**

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## 5. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

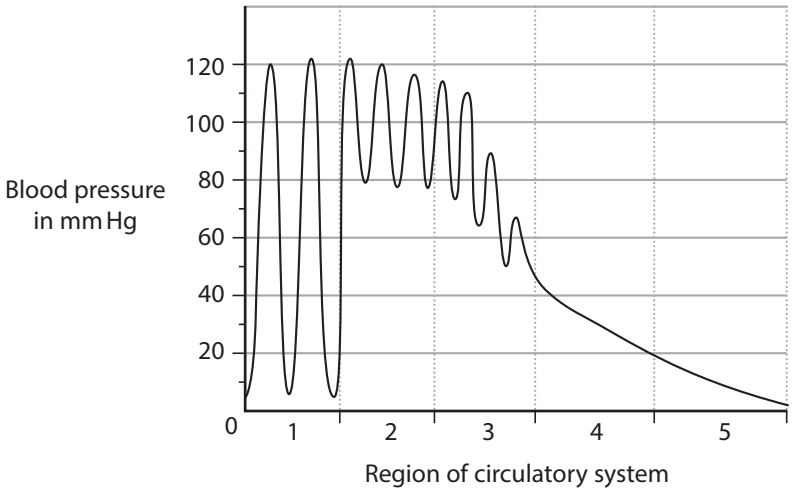
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## 6. Question 3 · part a.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part a.i only.

**3** The graph shows how blood pressure varies in different regions of the circulatory system.




(a) (i) Estimate the difference between the maximum blood pressure and minimum blood pressure in region 1. **(2)**

difference in blood pressure = ..... mm Hg

(ii) Which part of the circulatory system is represented by region 2? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

(iii) Which part of the circulatory system is represented by region 5? **(1)**

- ☒ **A** arteries
- ☒ **B** capillaries
- ☒ **C** heart
- ☒ **D** veins

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P 7 2 5 8 6 A 0 6 2 0

## 6. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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# 7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

**4** The eye contains receptors that can detect light.

(a) Name two other senses.

**(2)**

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

| Thickness of lens in mm | Focal length in mm |
|---|---|
| 10 | 12.5 |
| 9 | 14.0 |
| 8 | 15.5 |
| 7 | 18.0 |
| 6 | 21.0 |
| 5 | 25.0 |

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(i) Describe the relationship between the thickness of the lens and its focal length.

**(2)**

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(ii) Give the effect on the diameter of the lens as its thickness increases.

**(1)**

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## 7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

**(4)**

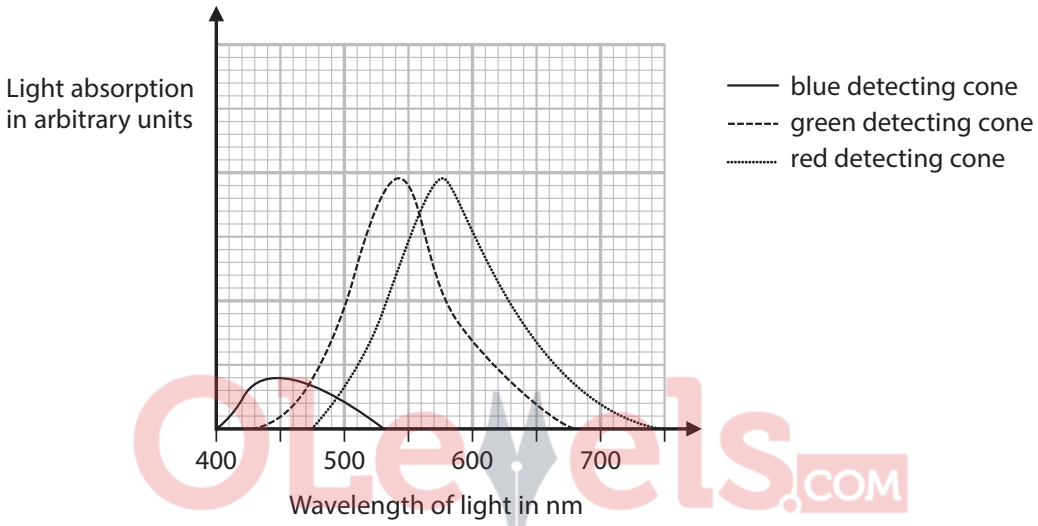
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# 7. Question 4 · part c.ii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.ii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.




(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

**(1)**

wavelength of light = ..... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

**(2)**

difference in wavelengths = ..... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

**(2)**

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**(Total for Question 4 = 14 marks)**

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## 7. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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# 8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

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**4** (a) (i) Complete the word equation for one type of respiration.

**(2)**

oxygen + ..... → ..... + .....

(ii) State the type of respiration shown by the equation.

**(1)**

.....

(iii) In which part of the cell does this type of respiration occur?

**(1)**

- ☒ **A** endoplasmic reticulum
- ☒ **B** mitochondrion
- ☒ **C** nucleus
- ☒ **D** ribosome

(iv) Name the molecule used to transfer energy in a cell.

**(1)**

.....

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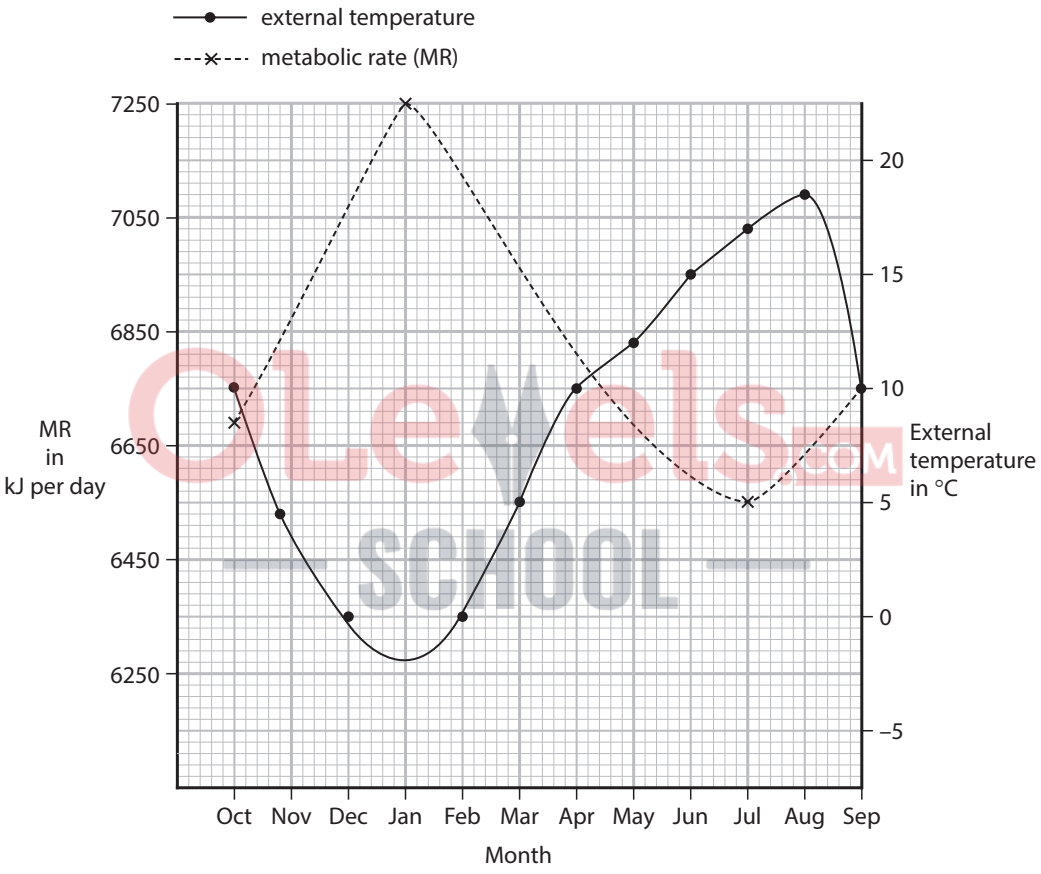
# 8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) An investigation was carried out to see how a person's metabolic rate (MR) and the external temperature change at different times in one year.

The graph shows the results of the investigation for one person.




(i) Describe how the investigation could be made more reliable.

**(2)**

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## 8. Question 4 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Determine the difference between the maximum metabolic rate and minimum metabolic rate.

**(2)**

difference = ..... kJ per day

(iii) Explain why the metabolic rate changes throughout the year.

**(5)**

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## 8. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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## 9. Question 3 · part b.i · 1 mark

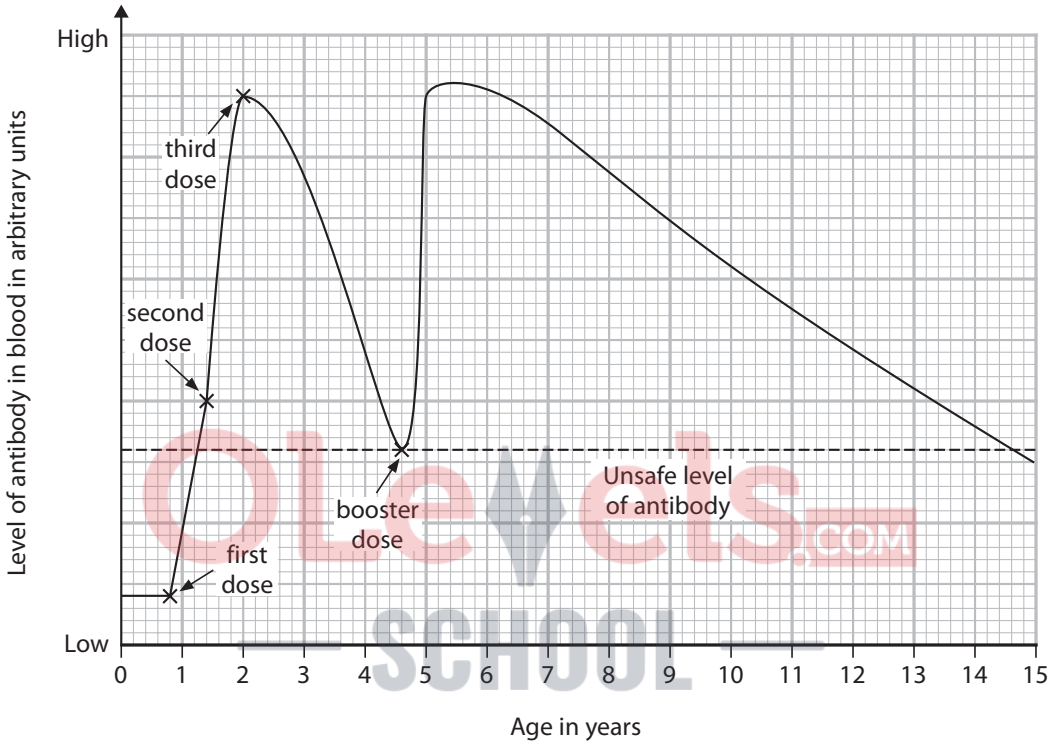
November 2023 Paper 2

Further practice · Attempt part b.i only.

**3** A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.




(a) State the general name given to disease causing organisms.

**(1)**

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

**(1)**

time = ..... years

(ii) State the type of immunity that vaccination produces.

**(2)**

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## 9. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

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# 10. Question 7 · part b.i · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

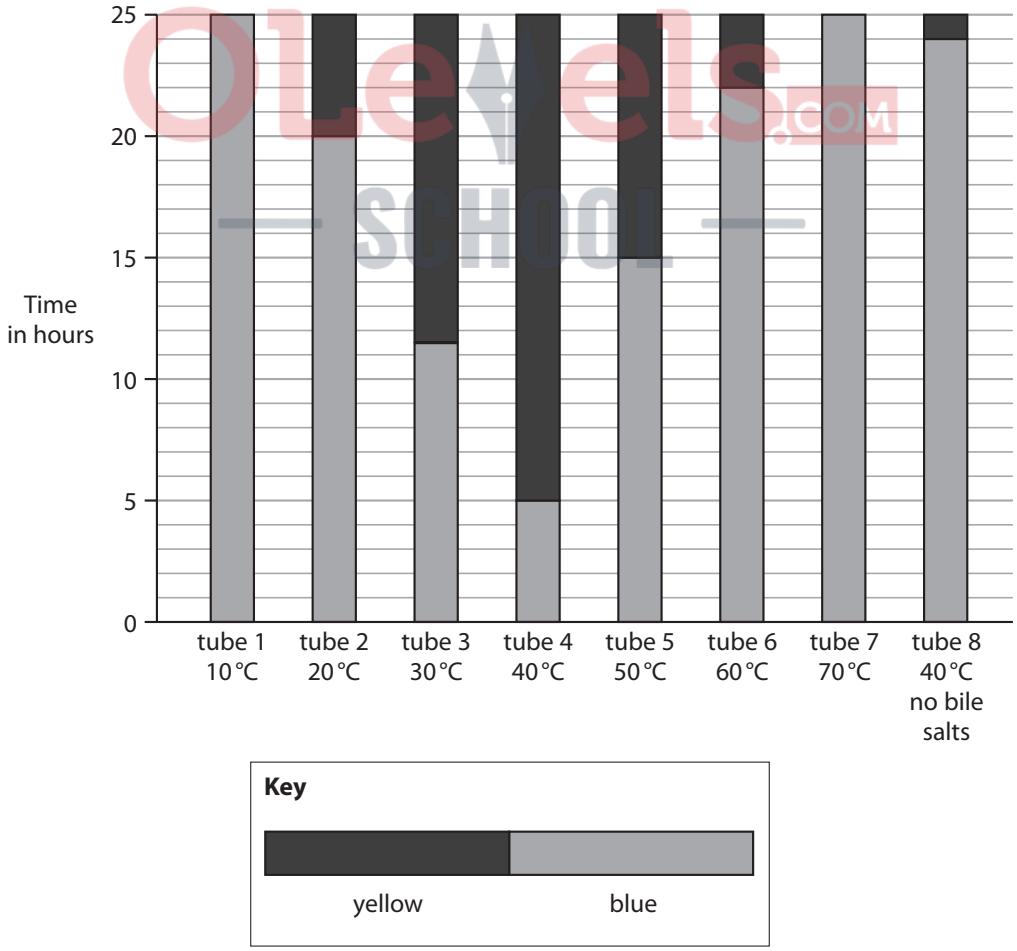
**7** A student investigates the digestion of olive oil (lipid) using the enzyme lipase, bile salts and a blue indicator.

The indicator changes from blue to yellow when the pH is below 5.

This is the student's method.

- mix together olive oil, lipase solution, bile salts and one drop of indicator
- put this mixture into a test tube
- repeat the process for six other test tubes
- keep the tubes at different temperatures
- keep another test tube (test tube 8) containing just olive oil, indicator and lipase at 40 °C
- record the colours of the mixture of each tube every hour for 25 hours

The chart shows the student's results.




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# 10. Question 7 · part b.i · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

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(a) (i) State three factors that should be kept constant in the student's investigation.

**(3)**

1 .....

2 .....

3 .....

(ii) State the optimum temperature for the action of lipase.

**(1)**

.....

(iii) Discuss why this temperature is only an approximate optimum temperature.

**(3)**

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(b) (i) Determine how much more time the indicator takes to change colour in tube 8 than in tube 3.

**(2)**

time = ..... hours

(ii) Explain why the pH decreases during the 25 hours.

**(3)**

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## 10. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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## 11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

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**1** (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

**(3)**

| Disease | Type of pathogen |
|---|---|
| cholera | |
| AIDS | |
| athlete's foot | |

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

**(4)**

| | | | |
|---|---|---|---|
| hormonal | antibiotics | contaminated water | virus |
| sexual intercourse | painkillers | bacterium | barrier |

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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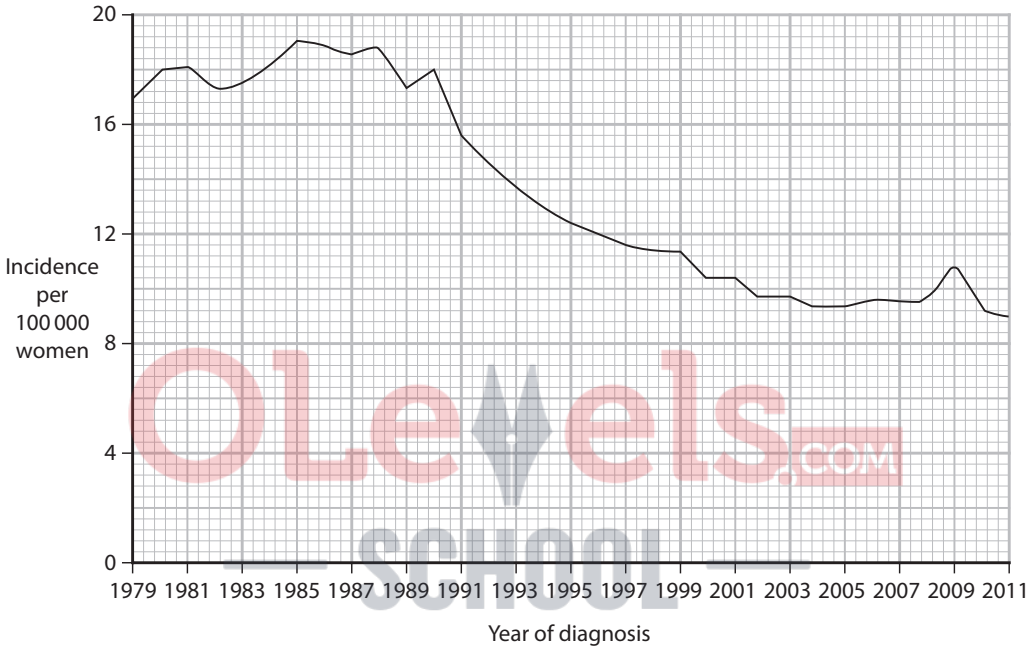
# 11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.




(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

**(2)**

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# 11. Question 1 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

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(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

**(2)**

difference = ..... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

**(1)**

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| | Contents of vaccination | Response by the body | How the pathogen is destroyed |
|---|---|---|---|
| ☒ **A** | antibodies | antigens produced | antigens destroy pathogen |
| ☒ **B** | antibodies | white blood cells produced | white blood cells destroy pathogen |
| ☒ **C** | antigens | antibodies produced | antibodies destroy pathogen |
| ☒ **D** | antigens | platelets produced | platelets destroy pathogen |

**(Total for Question 1 = 12 marks)**

Working and answer

# 12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

**2** To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

| Time in minutes | Blood glucose levels in mg per 100 $cm^3$ | |
|---|---|---|
| | person without diabetes | person with diabetes |
| 0 | 65 | 110 |
| 30 | 120 | 215 |
| 60 | 130 | 230 |
| 90 | 95 | 200 |
| 120 | 80 | 190 |
| 150 | 65 | 180 |
| 180 | 65 | 170 |

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## 12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

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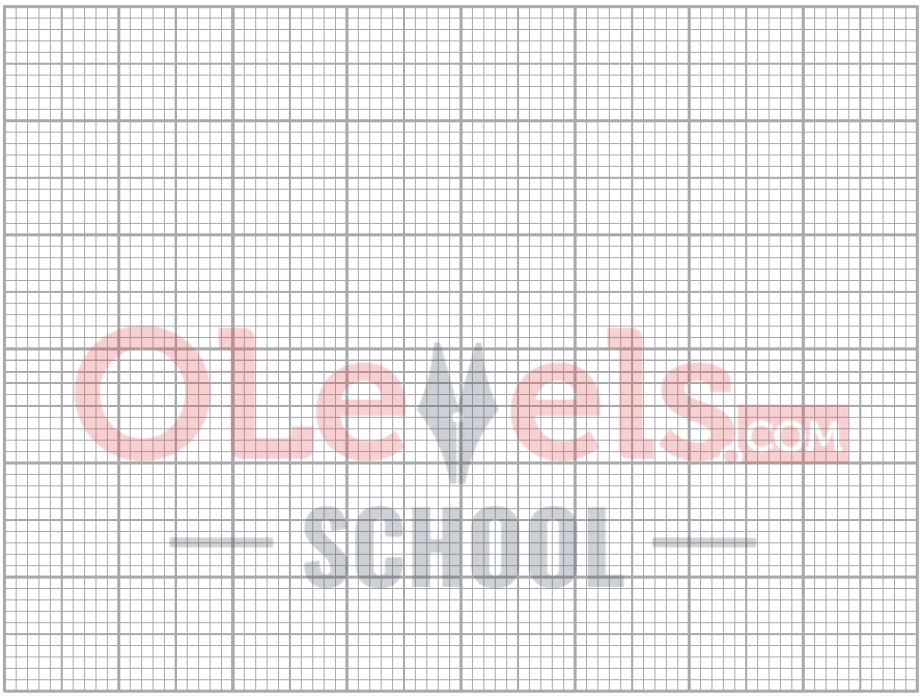
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(a) (i) Plot the results on the grid.

Join the points with straight lines.

**(6)**



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

**(2)**

without diabetes = ..... mg per 100 $cm^3$

with diabetes = ..... mg per 100 $cm^3$

## 12. Question 2 · part a.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.ii only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

**(3)**

1 .....

.....

2 .....

.....

3 .....

.....

(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

**(5)**

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**(Total for Question 2 = 16 marks)**

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## 12. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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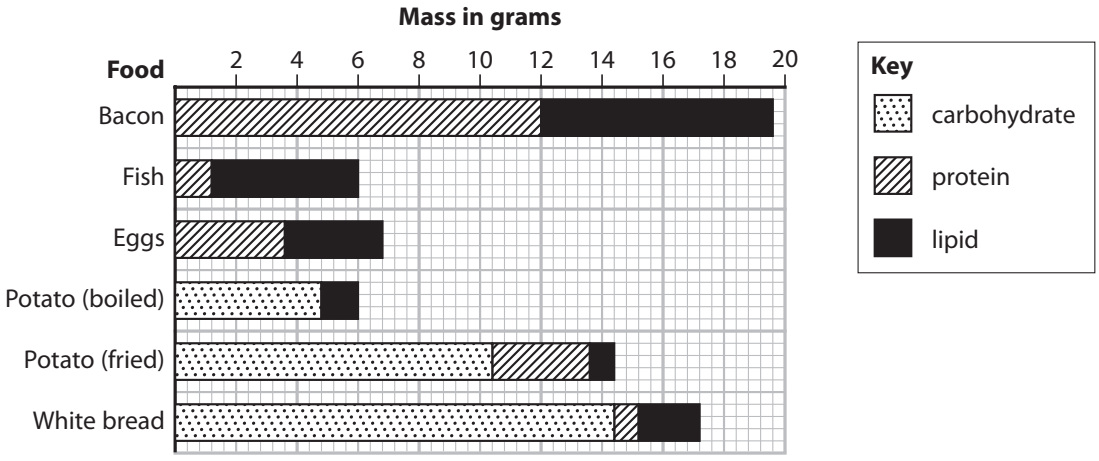
## 13. Question 7 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

**7** Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.




(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

**(1)**

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

**(1)**

(iii) Deduce the mass of lipid in 26 g of bacon.

**(2)**

mass of lipid = ..... g

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(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

**(3)**

total energy = ..... kJ

(b) Describe how to test for the presence of lipid in the fish.

**(3)**

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**(Total for Question 7 = 10 marks)**

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## 13. Question 7 · part a.iii · 2 marks · Working and answer

Working and answer

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